

PHASE 2

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# Requirement Analysis

Translating the ideation concept into functional, non-functional, and data requirements

## PulseView – Heart Disease Risk Analysis Dashboard

|                    |                                                   |
|--------------------|---------------------------------------------------|
| <b>Project</b>     | PulseView – Heart Disease Risk Analysis Dashboard |
| <b>Document</b>    | Requirement Analysis Report                       |
| <b>Version</b>     | 1.0                                               |
| <b>Date</b>        | July 2026                                         |
| <b>Prepared By</b> | Project Team                                      |

## 1. Introduction

This document translates the concept agreed upon during ideation into a concrete set of requirements. It covers what the system must do (functional requirements), the qualities it must have (non-functional requirements), the data it depends on, and the user stories that connect requirements back to the three personas identified earlier.

## 2. Functional Requirements

| ID   | Requirement             | Description                                                                                                                                                               |
|------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-1 | Home page               | Display a landing/hero page introducing PulseView and linking to the dashboard and story.                                                                                 |
| FR-2 | About page              | Present the problem statement, the three user personas, and a summary of the build process.                                                                               |
| FR-3 | Dashboard embed         | Embed the published Tableau dashboard with four supporting panels: Gender vs. Heart Disease, Diabetic vs. Stroke, Race-wise Heart Disease, and Smoking & Alcohol Impact.  |
| FR-4 | Guided story            | Embed the four-scene Tableau story (Overview, Gender Risk, Lifestyle Factors, Race & Conditions) with tab navigation, previous/next controls, and keyboard arrow support. |
| FR-5 | Contact form            | Accept name, email, and message fields; validate that none are empty; show a success or error flash message on submit.                                                    |
| FR-6 | Content configurability | Story scenes and dashboard panel copy must be data-driven (defined once in the application) so new content can be added without editing page templates.                   |

## 3. Non-Functional Requirements

| ID    | Category        | Requirement                                                                                                                                    |
|-------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-1 | Usability       | Every page must be understandable without any Tableau or statistics background.                                                                |
| NFR-2 | Performance     | Pages outside the embedded views should render in well under a second on a typical broadband connection.                                       |
| NFR-3 | Compatibility   | The site must render correctly in current versions of Chrome, Firefox, Edge, and Safari.                                                       |
| NFR-4 | Responsiveness  | Layout must adapt cleanly from desktop widths down to tablet and mobile widths.                                                                |
| NFR-5 | Maintainability | Adding a new story scene or dashboard panel should require only a data change in app.py, not a template edit.                                  |
| NFR-6 | Availability    | Dashboard and story availability depend on Tableau Public's uptime; the Flask app itself has no external service dependency beyond that embed. |

## 4. Data Requirements

The dashboard and story are built from a heart-disease risk-factor dataset combining demographic, lifestyle, and clinical fields. The following attributes are referenced directly by the dashboard panels, the story scenes, or the persona use cases:

| Field                                 | Used in                                                                     |
|---------------------------------------|-----------------------------------------------------------------------------|
| Heart disease status (outcome)        | All dashboard panels; all four story scenes                                 |
| Sex                                   | Gender Risk story scene; Gender vs. Heart Disease panel                     |
| Smoking status                        | Lifestyle Factors scene; Smoking & Alcohol Impact panel                     |
| Alcohol consumption                   | Lifestyle Factors scene; Smoking & Alcohol Impact panel                     |
| Race                                  | Race & Conditions scene; Race-wise Heart Disease panel                      |
| Diabetes history                      | Race & Conditions scene; Diabetic vs. Stroke panel                          |
| Stroke history                        | Race & Conditions scene; Diabetic vs. Stroke panel                          |
| Age, BMI, cholesterol, blood pressure | Referenced by the cardiologist and patient personas for deeper segmentation |

## 5. User Stories

| As a...                   | I want to...                                                                      | So that...                                                              |
|---------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Cardiologist (Dr. Sharma) | filter the dashboard by age, BMI, cholesterol, and smoking status                 | I can identify high-risk patient groups and design targeted campaigns.  |
| Policy officer (Ramesh)   | compare heart-disease prevalence and sedentary-lifestyle indicators across groups | I can direct prevention budget to where it will have the most impact.   |
| Patient (Anita)           | see a simple visual comparison of my own risk factors against benchmarks          | I can understand my personal risk without needing a medical background. |
| First-time visitor        | read a short guided story before the full dashboard                               | I get context on what the data means before exploring it myself.        |
| Any visitor               | reach out through a contact form                                                  | I can ask questions about the project or the data.                      |

## 6. Technology & Tooling Requirements

- **Backend:** Python 3, Flask 3.1.3
- **Visualization:** Tableau (workbook authoring) and Tableau Public (hosting the interactive embeds)
- **Frontend:** Jinja2 templates, hand-written CSS, vanilla JavaScript (no frontend framework)
- **Environment:** Python virtual environment (venv) for dependency isolation

## 7. Assumptions & Constraints

- The Tableau workbook is published and publicly accessible on Tableau Public; the Flask app has no control over its rendering performance.
- The contact form captures messages but has no email backend wired up yet — this is a known, deliberate scope boundary for this iteration.
- No user authentication or personal data storage is in scope; all dashboard data is aggregate, not individual patient records.